

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-5

Faculty

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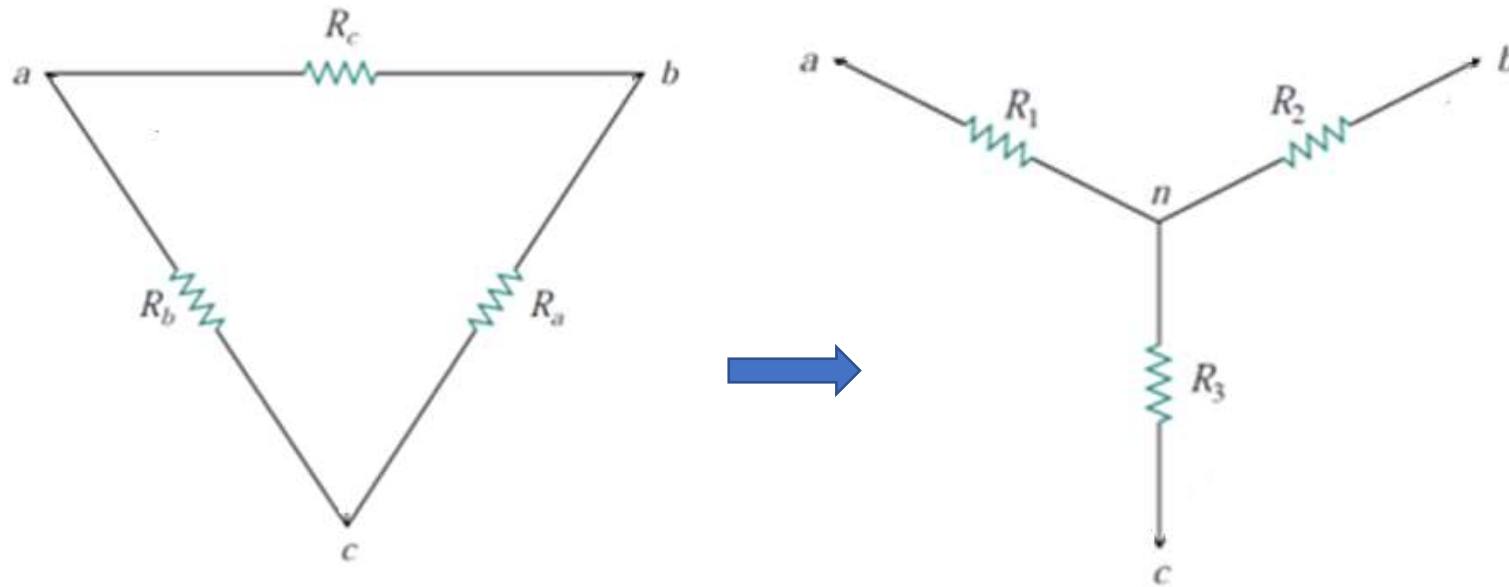
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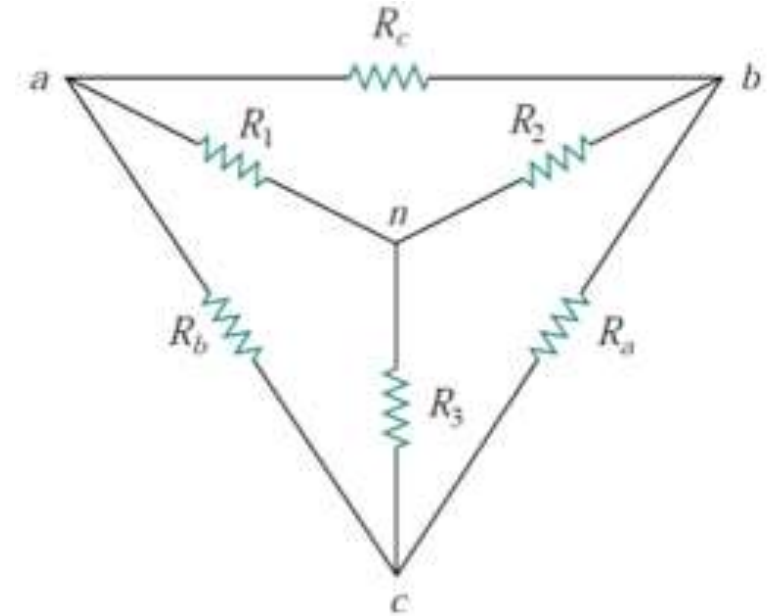
Cabin: Faculty Area 424-C:AB-1

1.DELTA(Δ) to STAR(Y)



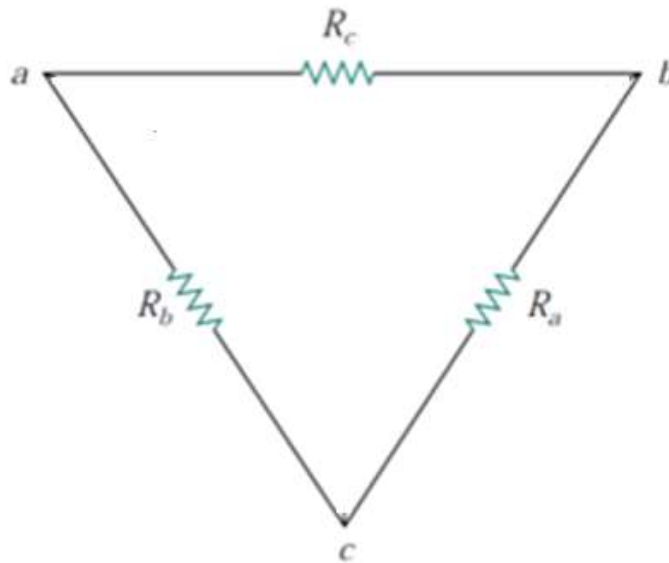
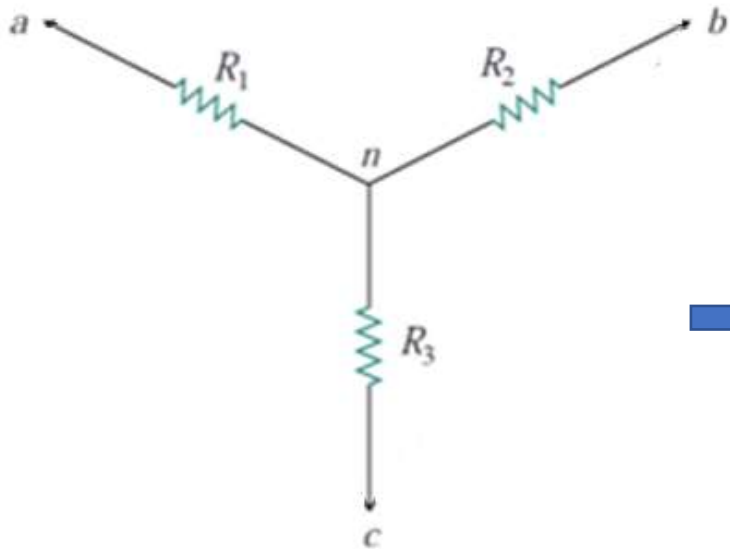
$$R_1 = \frac{(R_b R_c)}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{(R_a R_c)}{(R_a + R_b + R_c)}$$



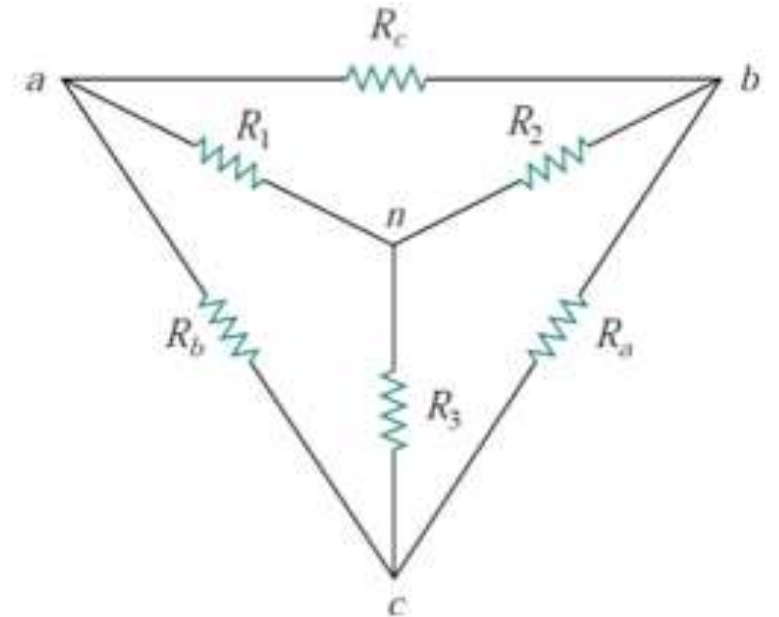
$$R_3 = \frac{(R_a R_b)}{(R_a + R_b + R_c)}$$

2.STAR(Y) to DELTA(Δ)



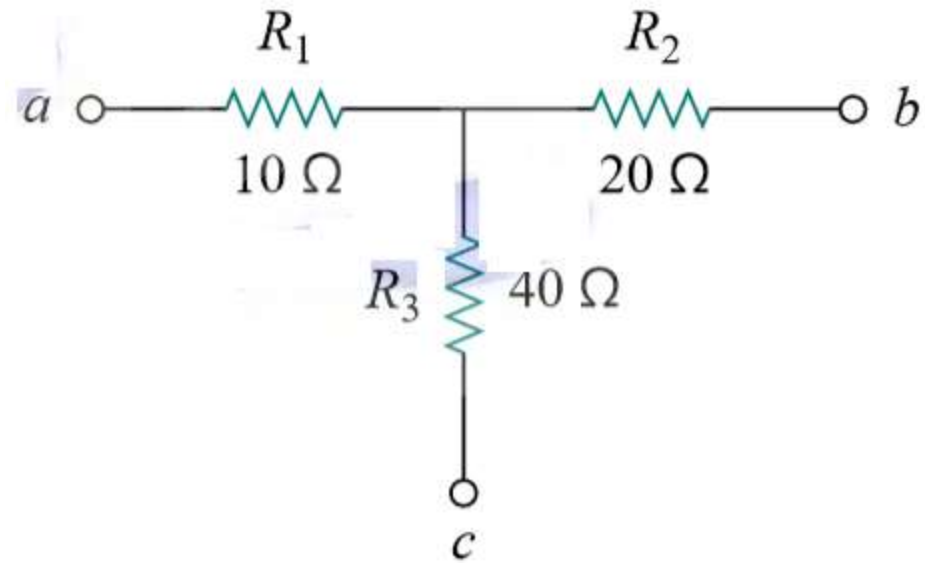
$$R_a = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_1}$$

$$R_b = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_2}$$

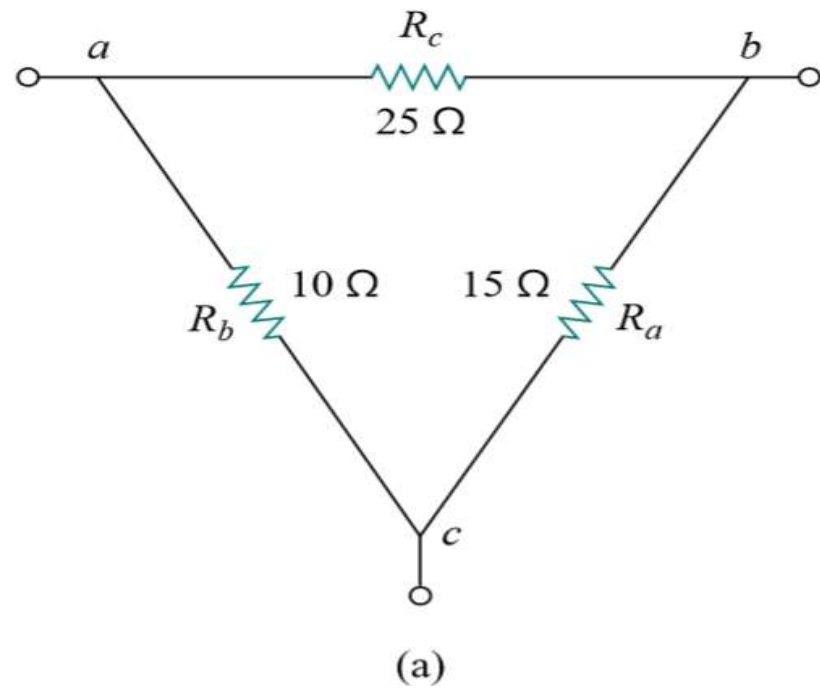


$$R_c = \frac{(R_1 R_2) + (R_2 R_3) + (R_3 R_1)}{R_3}$$

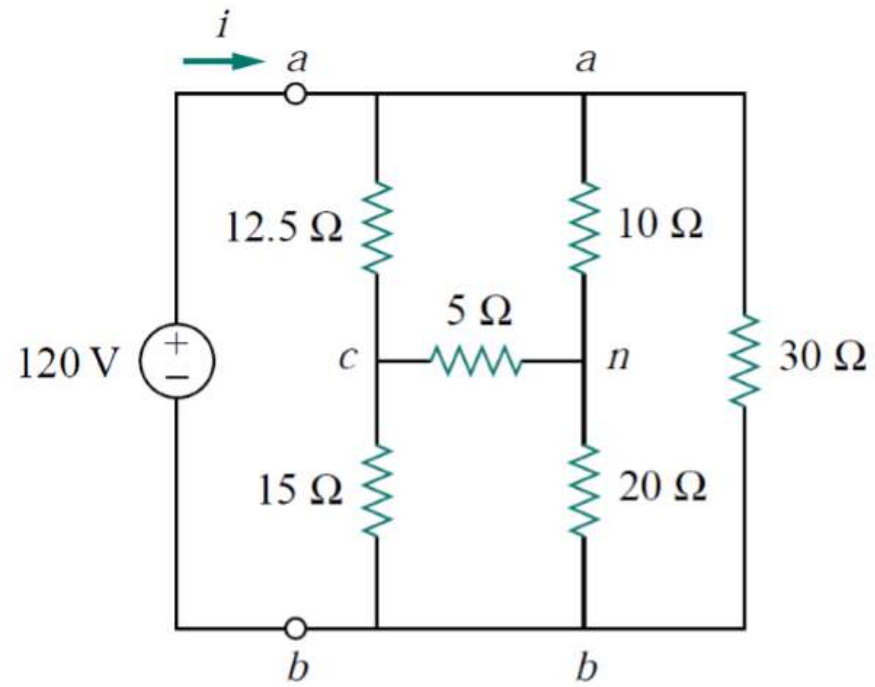
Problem-1: Convert Star to Delt Network



Problem-2: Convert Delt to Star Network

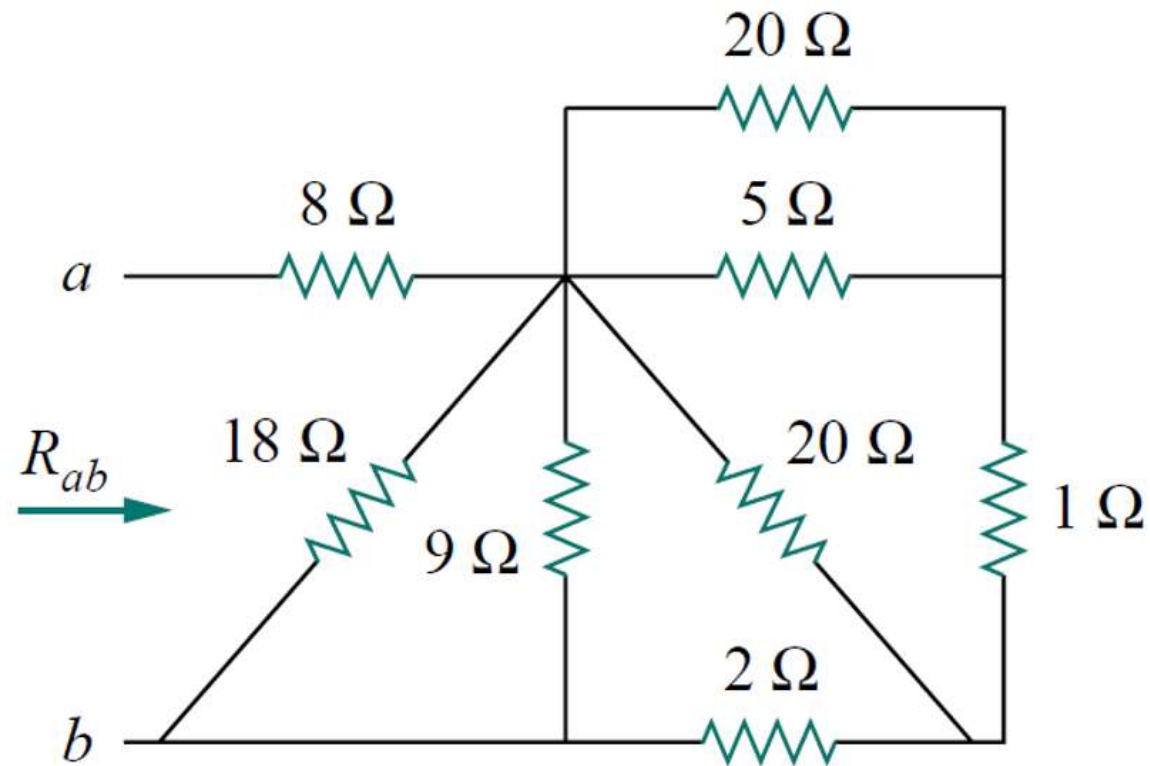


Problem-3: Find R_{ab}



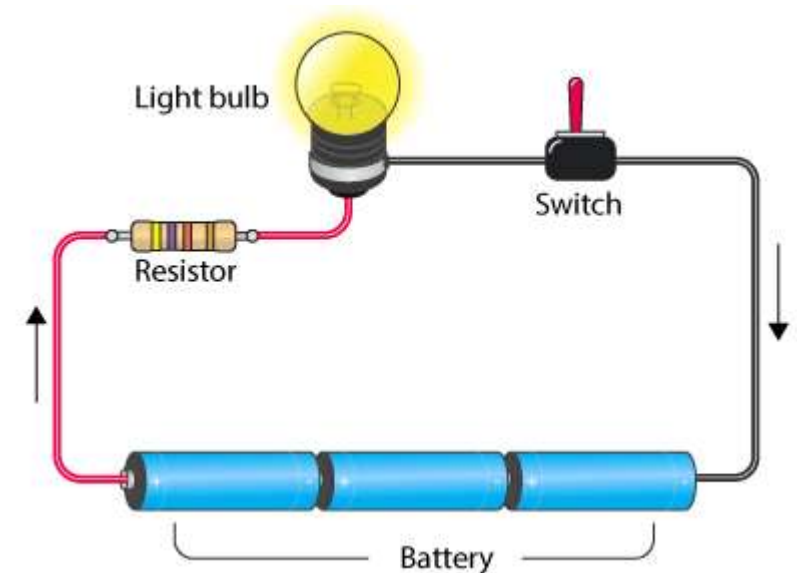
Problem-4

Find the R_{ab}



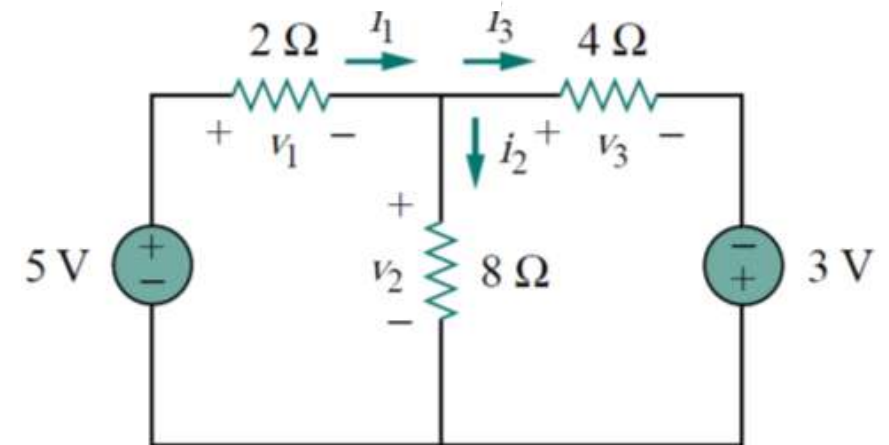
I. Ohm's Law

- Ohm's law states that the voltage V across a resistor is directly proportional to the current I flowing through the resistor, provided there is no change in the physical conditions.
 - $V \propto I$.
 - Resistance (R) is the proportionality constant. $V = I R$.
 - Also $I = G V$, where G is conductance or inverse of resistance.
- Electrical power (P or p):
 - $P = V \times I = I^2 \times R = V^2 / R$
- Electrical energy (E) :
 - $E = V \times I \times t = I^2 \times R \times t = [V^2 / R] \times t$



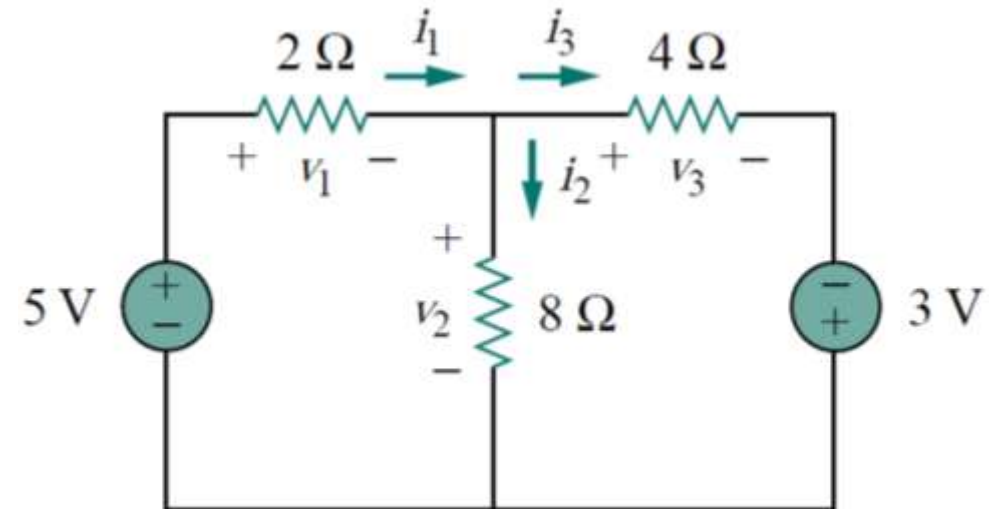
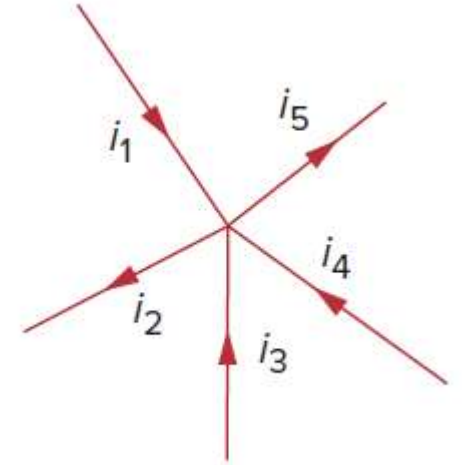
Kirchhoff's Laws

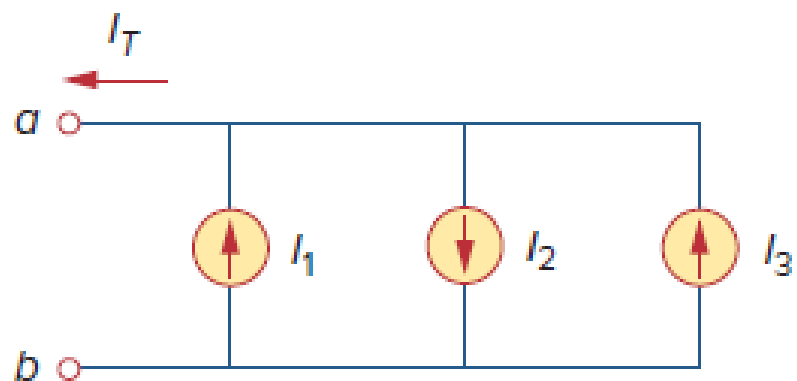
- Ohm's law by itself is not sufficient to analyze circuits.
- However, when it is coupled with Kirchhoff's two laws, we have a sufficient, powerful set of tools for analyzing a large variety of electric circuits.
- Kirchhoff's laws were first introduced in 1847 by the German physicist Gustav Robert Kirchhoff (1824–1887).
- These laws are formally known as Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL).



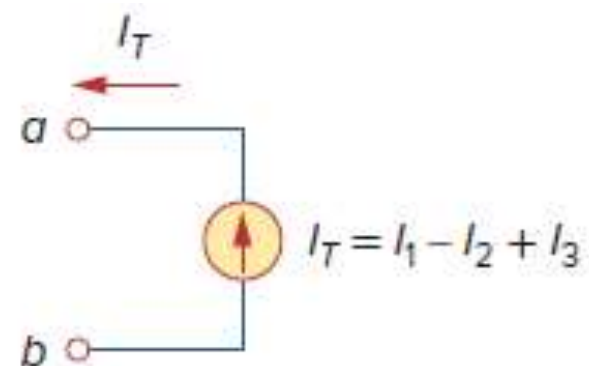
1. Kirchhoff's Current Law (KCL)

- Kirchhoff's first law is based on the law of conservation of charge, which requires that the algebraic sum of charges within a system cannot change.
- Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.
- Mathematically, KCL implies that $\sum I = 0$, in a node.
- By this law, currents entering a node may be regarded as positive, while currents leaving the node may be taken as negative or vice versa.
- Alternatively, KCL can be stated as : The sum of the currents entering a node is equal to the sum of the currents leaving the node.



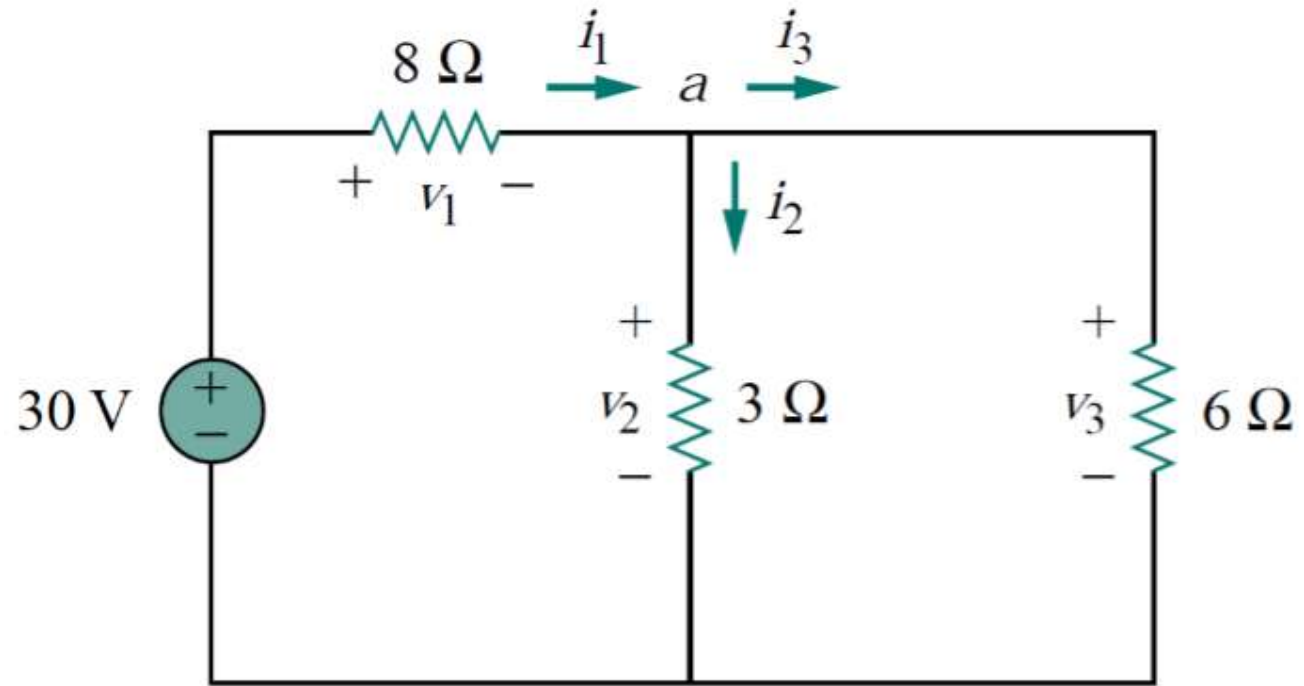


(a)



(b)

Problem-1: Find all branch currents?



Cramer's Rule: (two variables)

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Let us write these two equations in the form of $AX = B$.

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$D = |A| = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$$

And

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = c_1b_2 - c_2b_1$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = a_1c_2 - a_2c_1$$

$$x = \frac{D_x}{D}$$

$$y = \frac{D_y}{D}$$

Cramer's Rule: (three variables)

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$x = \frac{D_x}{D}, y = \frac{D_y}{D}, z = \frac{D_z}{D}, D \neq 0$$

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

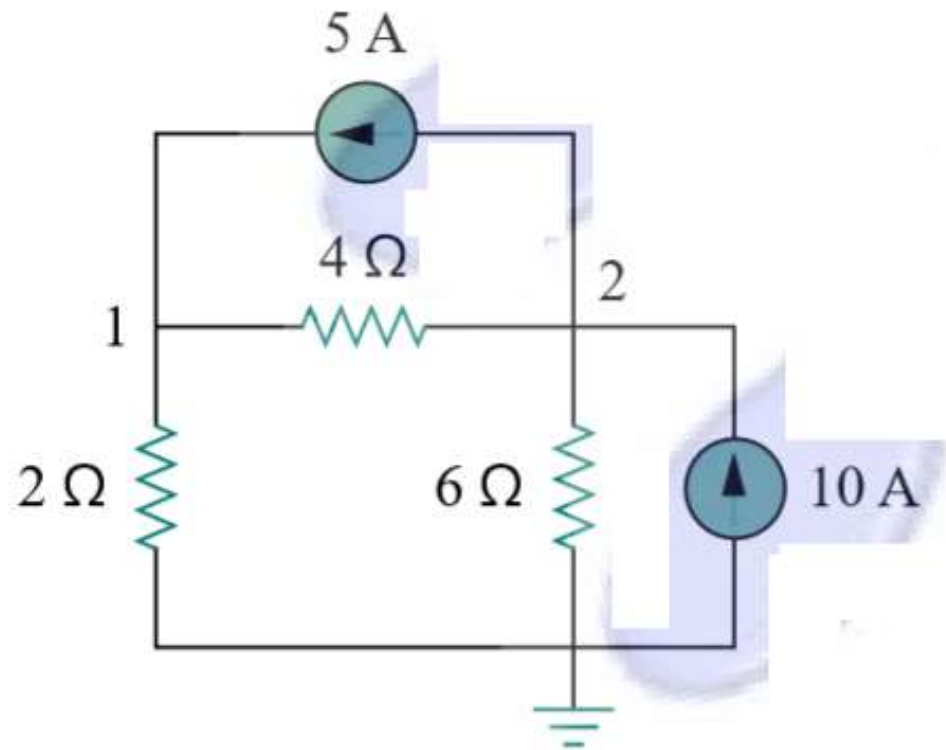
$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} a_1 & b_1 & c_1 & a_1 & b_1 \\ a_2 & b_2 & c_2 & a_2 & b_2 \\ a_3 & b_3 & c_3 & a_3 & b_3 \end{vmatrix}$$

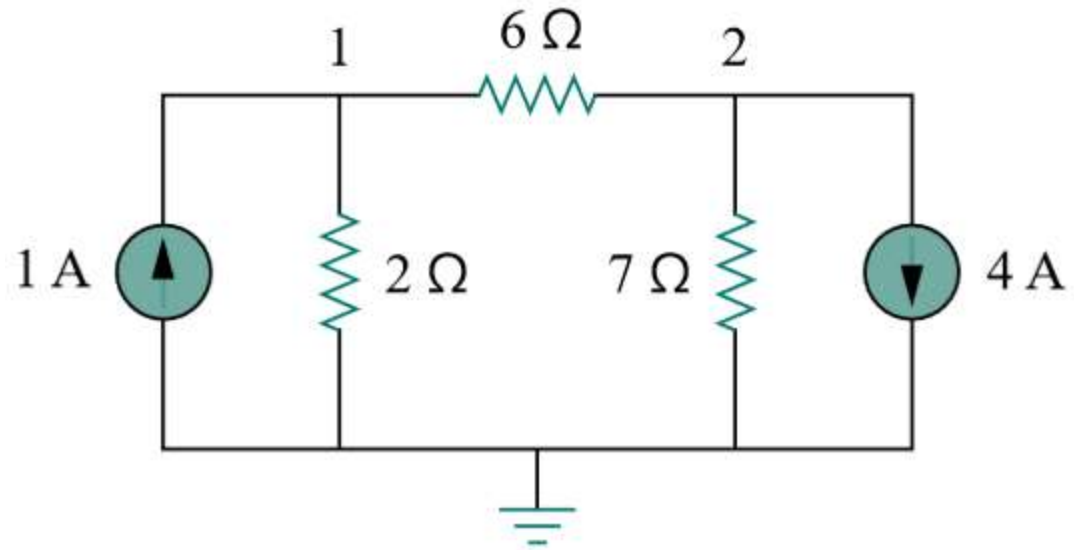
$$\det(A) = \begin{vmatrix} a_1 & b_1 & c_1 & a_1 & b_1 \\ a_2 & b_2 & c_2 & a_2 & b_2 \\ a_3 & b_3 & c_3 & a_3 & b_3 \end{vmatrix}$$

$$|A| = a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3 - a_3b_2c_1 - b_3c_2a_1 - c_3a_2b_1$$

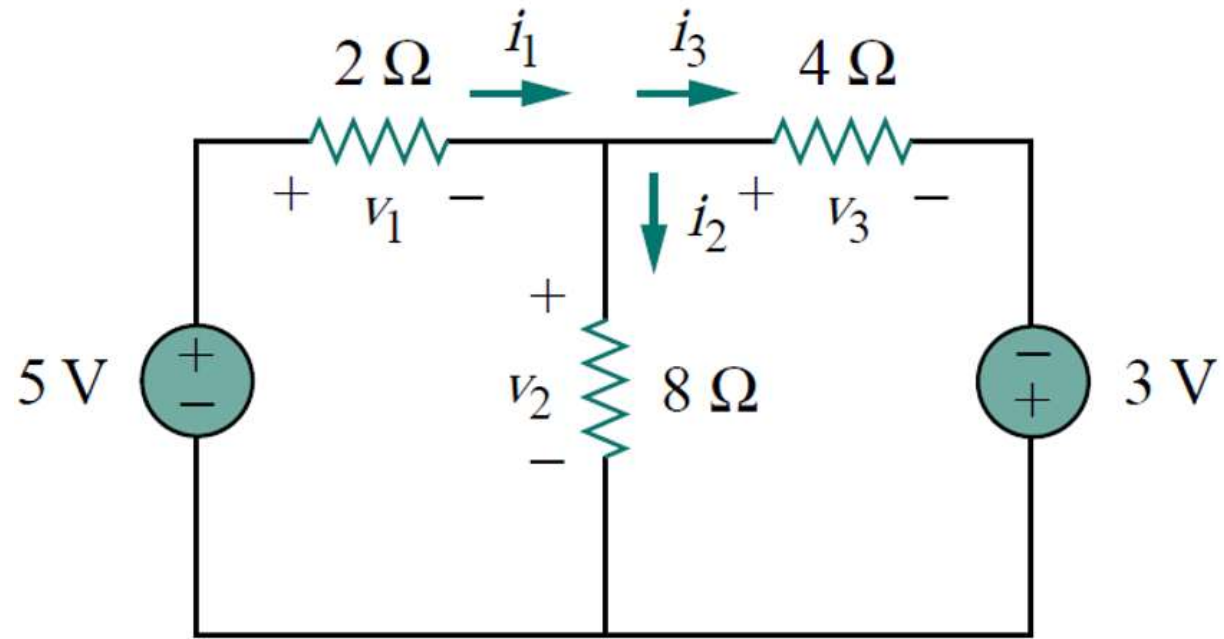
Problem: Find all branch currents?



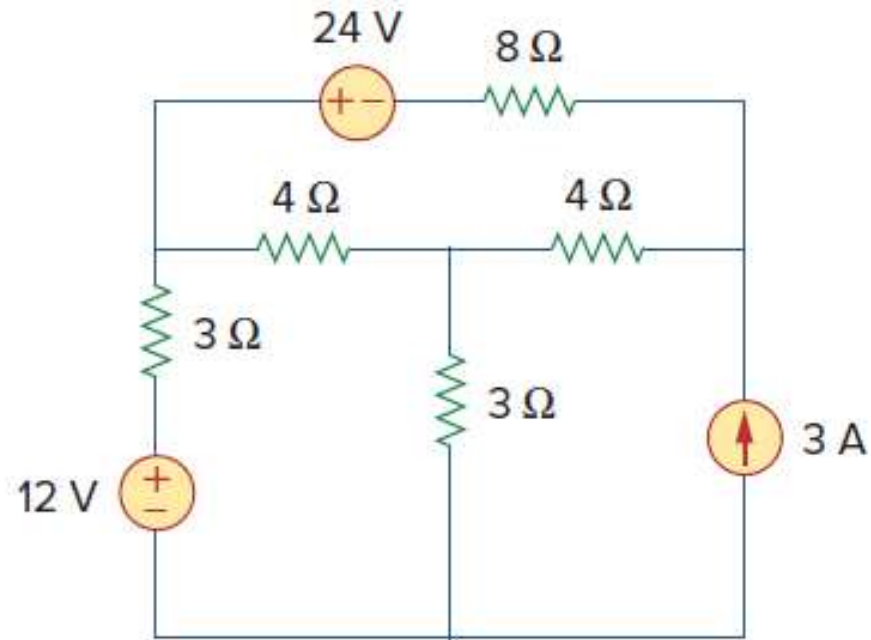
Problem: Find all branch currents?



Problem: Find all branch currents?



Problem: Find all branch currents?



Problem : find “ i ” in the circuit

